

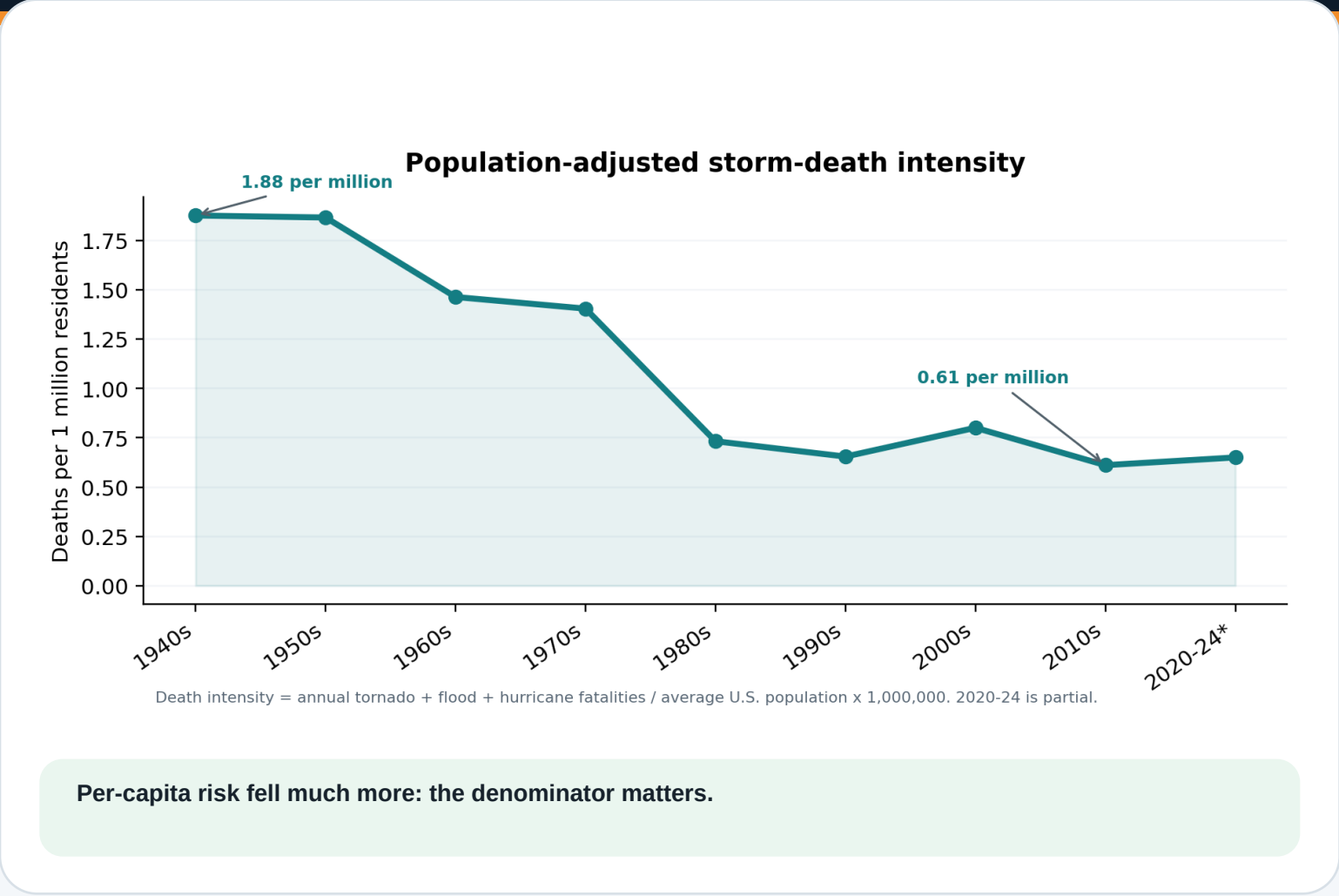
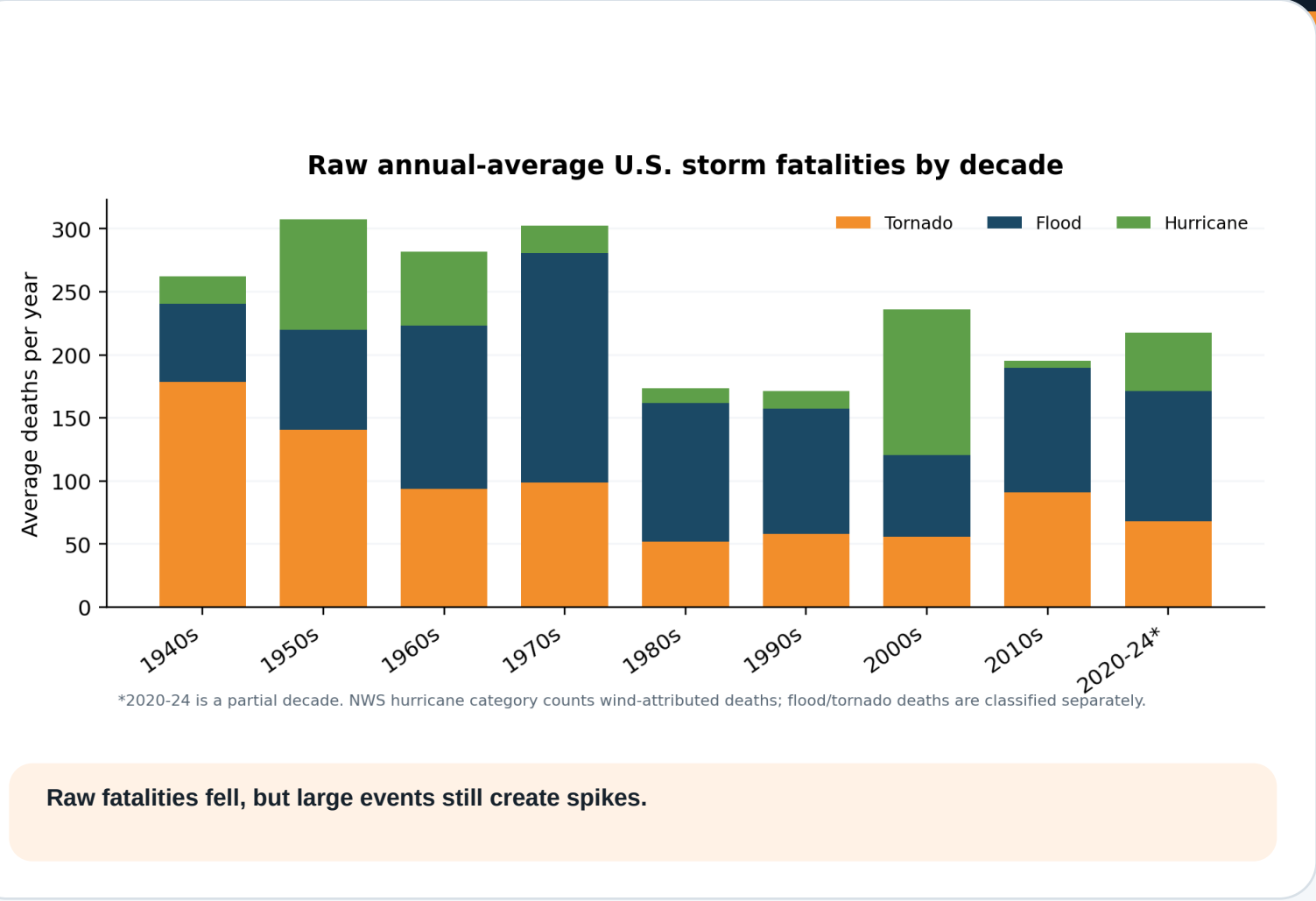
The Denominator Changes the Story

U.S. extreme-weather risk is not just a story about storms. It is also a story about population, exposure, vulnerability, and adaptation.

Risk = Hazard x Exposure
Vulnerability and adaptation shape what exposure turns into.

-26%
raw storm deaths
1940s to 2010s

-67%
death intensity
1940s to 2010s



2. Exposure

People do not only flee hazards.
Maricopa County gained 46,866 domestic migrants in 2020-21 and 33,305 in 2021-22. That is movement within the U.S., not births or international immigration.

Maricopa County, AZ: people moved toward heat exposure

Period	Net domestic migrants
2020-21	46,866
2021-22	33,305

Census Bureau net domestic migration; this excludes births and international migration.

Phoenix heat exposure:
111 days per year at or above 100 F (1991-2020 average).

3. Heat & cold

The metric matters.
NWS heat data begin in 1986 and show large spikes, including 1,021 direct heat fatalities in 1995. But Burke et al. (2025) estimate something different:

Burke et al. finding
Across 30 countries, ambient temperature is associated with about 5-12% of total deaths; cold exceeds heat in every context they study.

IMPORTANT DISTINCTION
Direct NWS weather-death counts are not the same thing as statistical temperature-attributable mortality. Do not compare the magnitudes as if they measure the same outcome.

What economics adds
The public argument often begins with the hazard: are storms or temperatures getting worse? Economics asks a broader question: what happens when hazards interact with where people live, what they own, and how they adapt?

People trade off risk against jobs, housing, amenities, energy costs, and other benefits. Institutions shape zoning, insurance, building codes, warnings, cooling access, and infrastructure. Technology and wealth can lower vulnerability - but not evenly for everyone.

- Population denominator**
Raw deaths fell, but deaths per million fell much more.
- Adaptation**
Forecasts, infrastructure, air conditioning, warnings, and behavior can reduce vulnerability.
- Exposure**
Migration and development can put more people and assets in risky places.
- Tradeoffs**
The useful policy question is which intervention buys the most risk reduction for its cost.

IPCC SREX: losses are not just the hazard
The IPCC SREX report concludes with high confidence that increasing exposure of people and economic assets has been the major cause of long-term increases in weather- and climate-related disaster losses. Vulnerability also matters.

Economic lesson: separate hazard, exposure, and adaptive capacity.

BUT WAIT: decline does not mean "problem solved"
The evidence supports a narrower conclusion: U.S. storm-mortality risk per person has fallen in these categories. It does not prove storms are harmless, dollar damages are falling, every group is equally protected, or future heat risk will be easy to manage.

Better question: what adaptation reduces risk most effectively?